CvSU Vision
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Republic of the Philippines
CAVITE STATE UNIVERSITY
Bacoor City Campus
SHIV, Molino VI, City of Bacoor

CvSU Mission
Cavite State University shall provide, excellent equitable and relevant education opportunities in the arts, science and technology through quality instruction and relevant research and development activities.
It shall produce professional, skilled and morally upright individuals for global competitiveness.

DEPARTMENT OF COMPUTER STUDIES
Bachelor of Science In Computer Science

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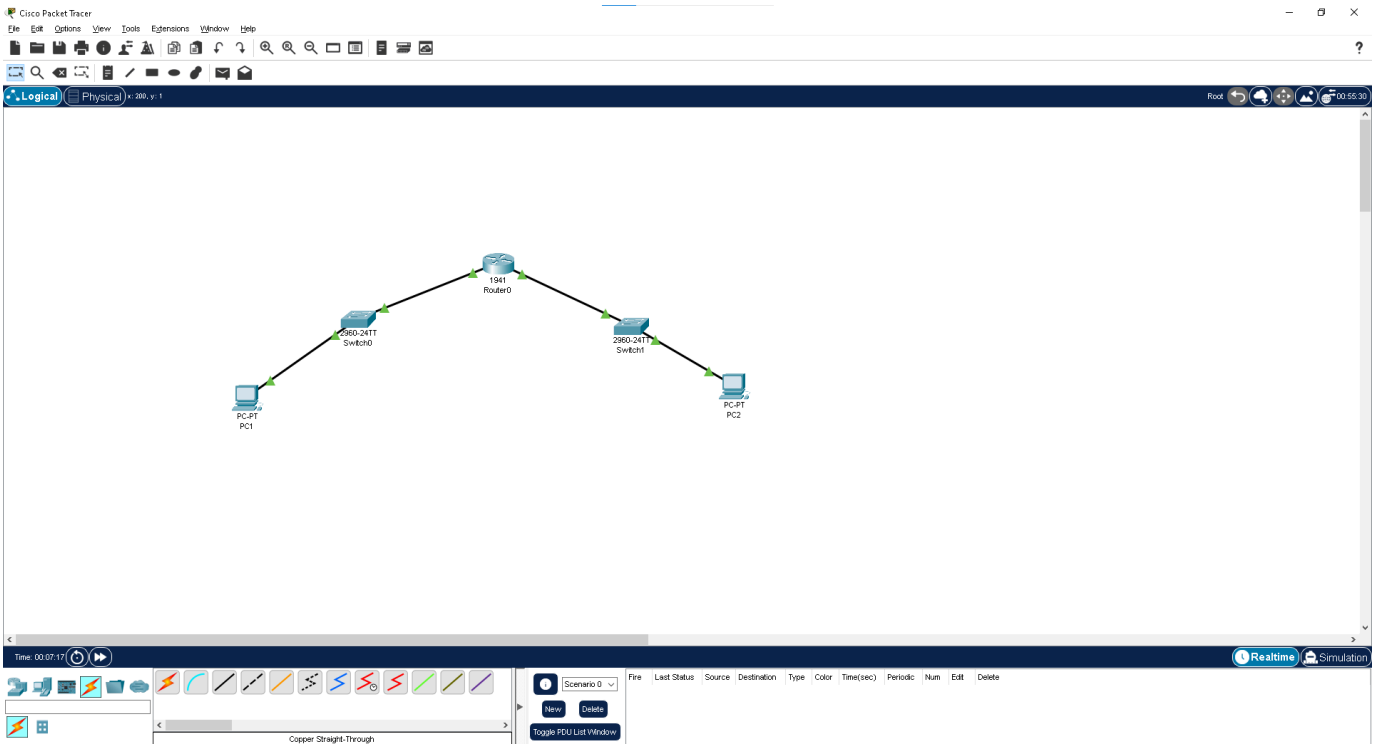
Year Level & Section:	**BSCS 3-5**	**Date:**	**10/24/25**

Laboratory Activity #3

Logical Network Topology



This shows the logical network topology where two PCs are connected through switches and a router. It illustrates how devices are logically linked to enable communication within the network.

Router Command Line Interface

Router0

Physical Config CLI Attributes

IOS Command Line Interface

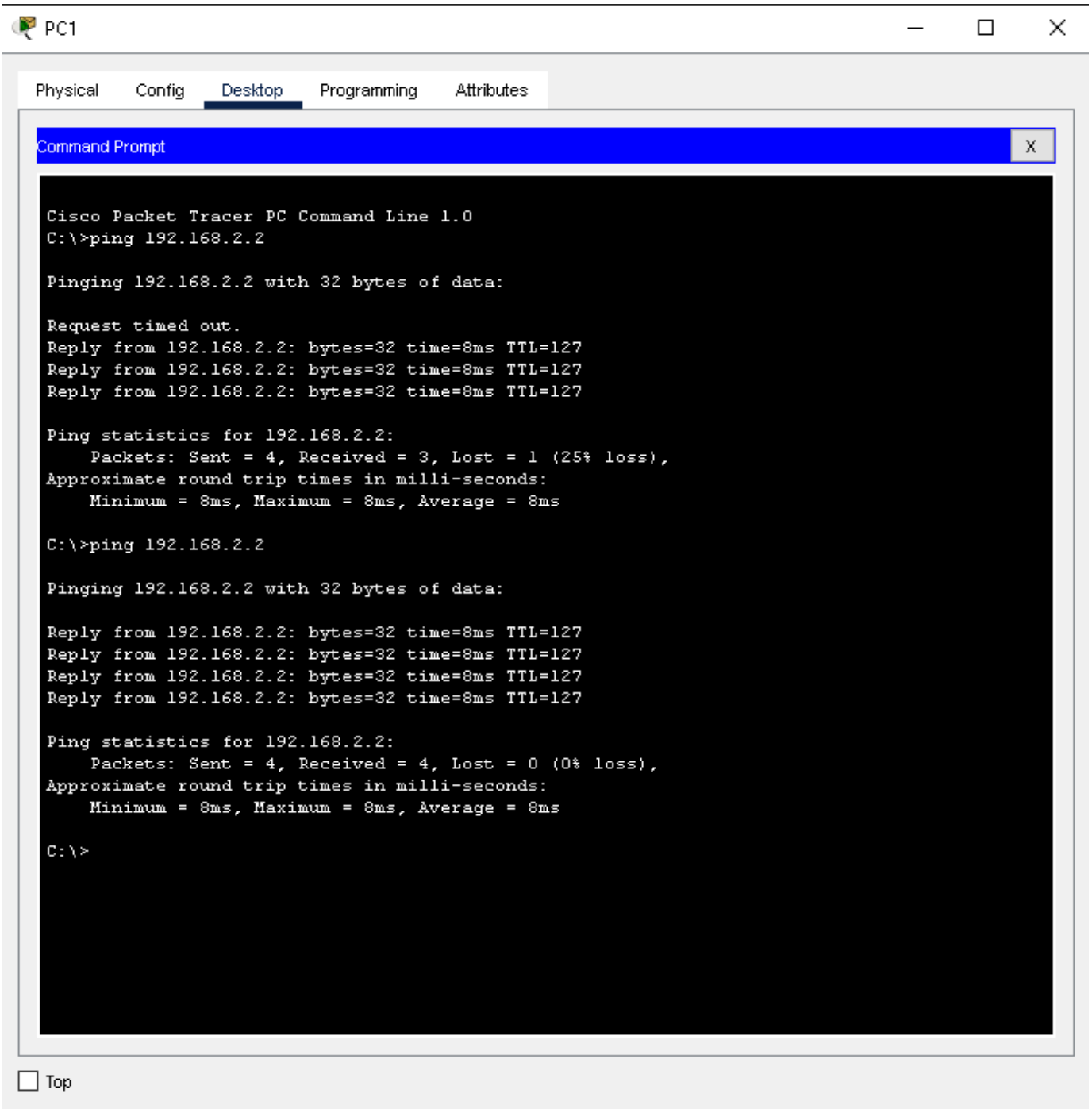
```
         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

Copy Paste

Top

This shows the router configuration process where IP addresses are assigned to GigabitEthernet interfaces. The “no shutdown” command activates each interface, allowing network communication between connected devices.

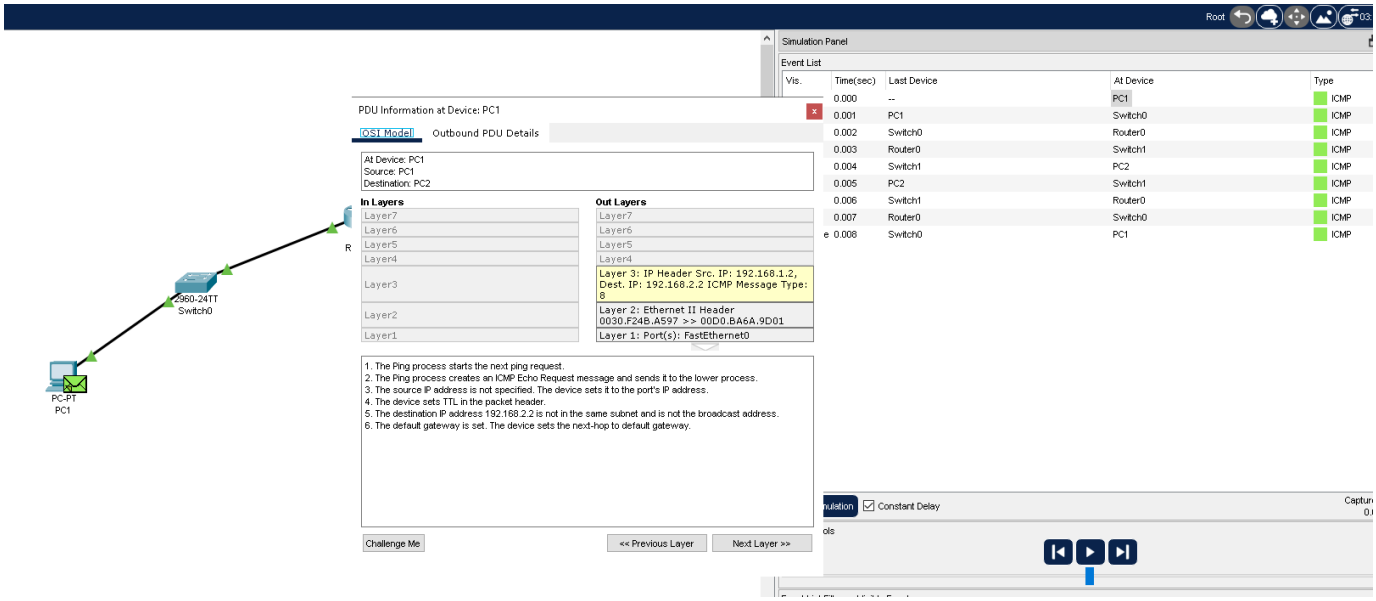
Ping From PC1 to PC2



This shows a successful ping from PC1 to PC2, indicating that both computers are properly configured and connected within the same network. The continuous replies with 0% packet loss confirm that the communication between the two devices is stable and the network connection is functioning correctly.

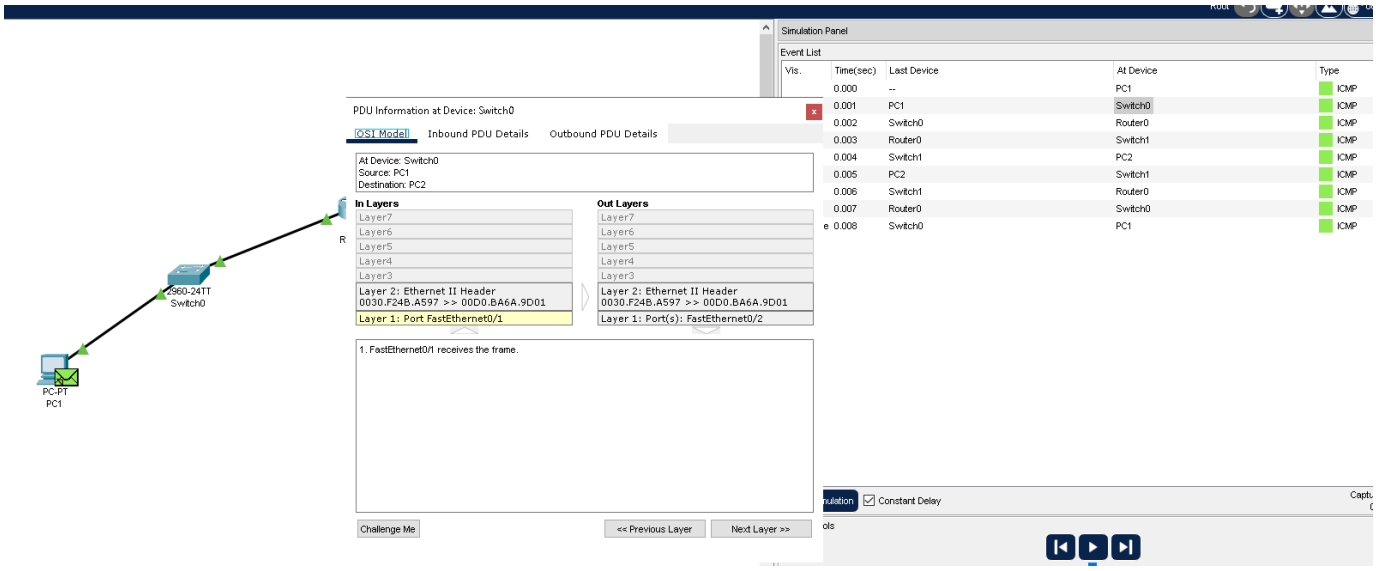
Encapsulation:

At PC1:



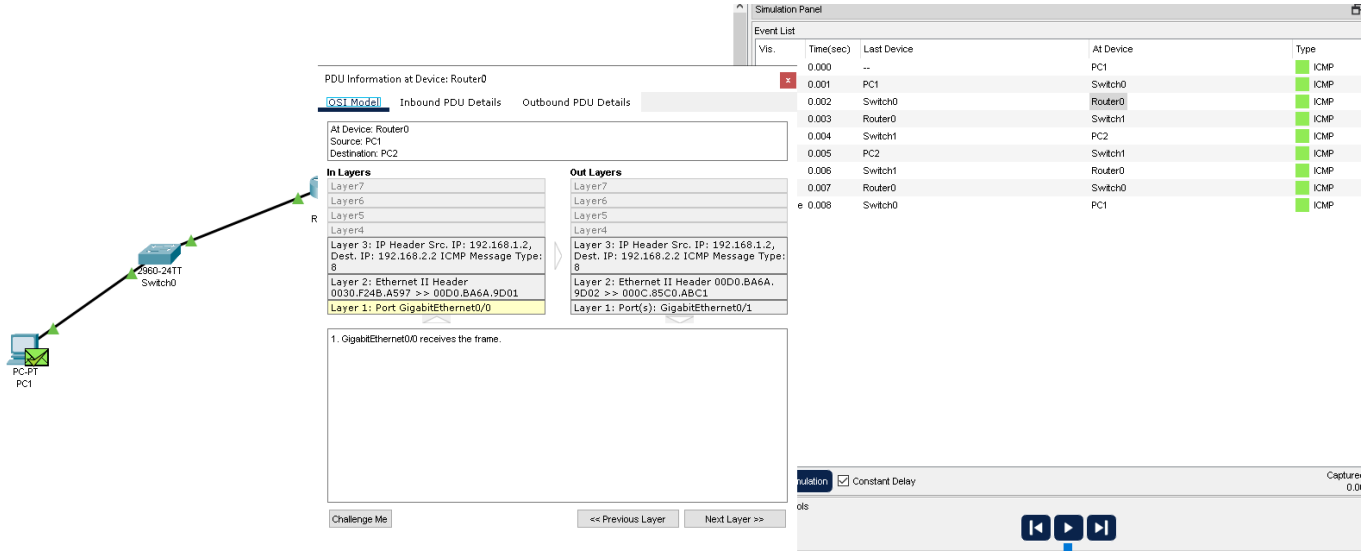
This part shows the start of encapsulation at PC1, where the device generates an ICMP Echo Request (ping) to PC2. The process mainly begins at the Network Layer (Layer 3) because ping uses the ICMP protocol, which operates at this layer. Then, the Data Link Layer (Layer 2) adds the Ethernet frame to prepare the packet for physical transmission.

From PC1 to Switch0:



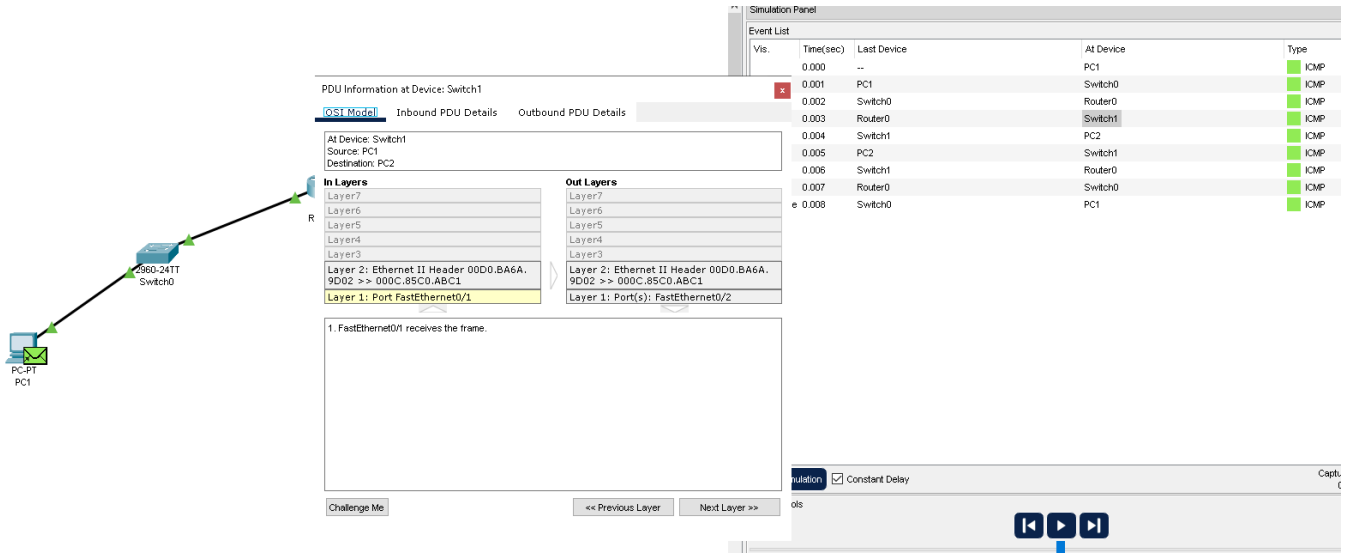
This part shows the frame being received and forwarded by the switch (Switch0). The switch operates at Layer 2 (Data Link Layer) of the OSI model. It reads the MAC addresses in the Ethernet frame header to determine where to forward the frame.

From Switch0 to Router:



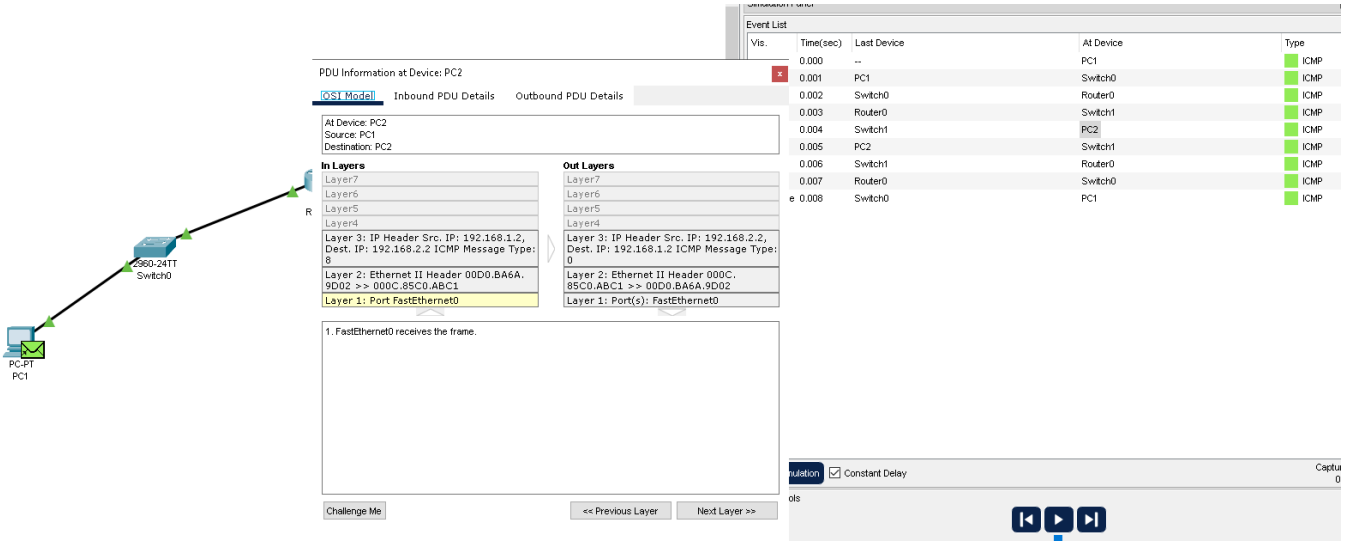
This step shows the router (Router0) receiving the frame from the switch and processing it at the Network Layer (Layer 3). The router removes the old Ethernet frame (Layer 2) and checks the IP header (Layer 3) to determine where to send the packet next.

From Router to Switch1:



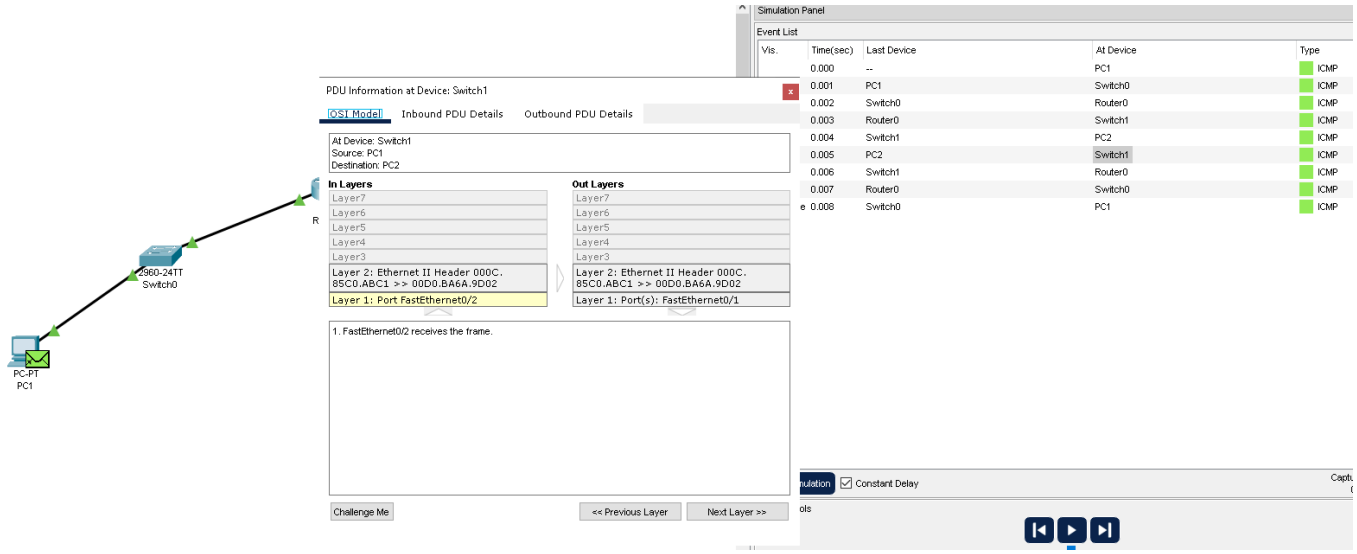
This step shows Switch1 receiving a data frame from the router through its FastEthernet0/1 interface. At this point, the switch is functioning primarily at the Data Link Layer (Layer 2) of the OSI model, which is responsible for handling physical addressing and frame forwarding.

From Switch1 to PC2:



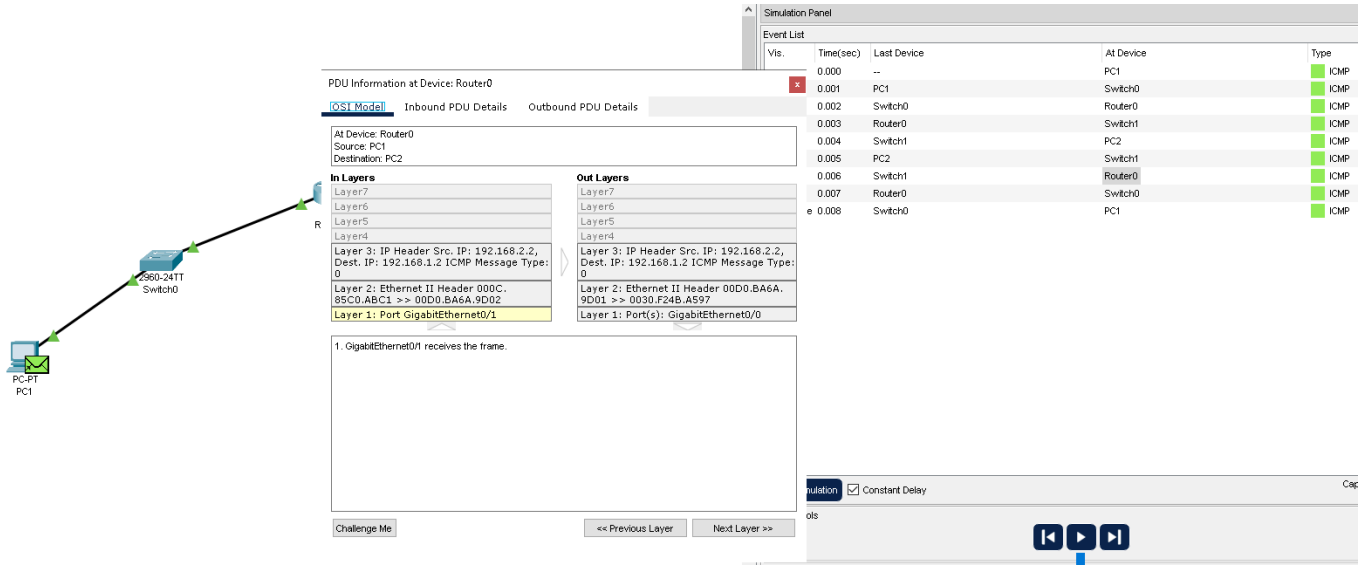
This step shows PC2 receiving the ICMP packet from PC1 through its FastEthernet0 interface. The message has successfully reached its destination, meaning the communication between both devices is working correctly.

Decapsulation:
From PC2 to Switch1:



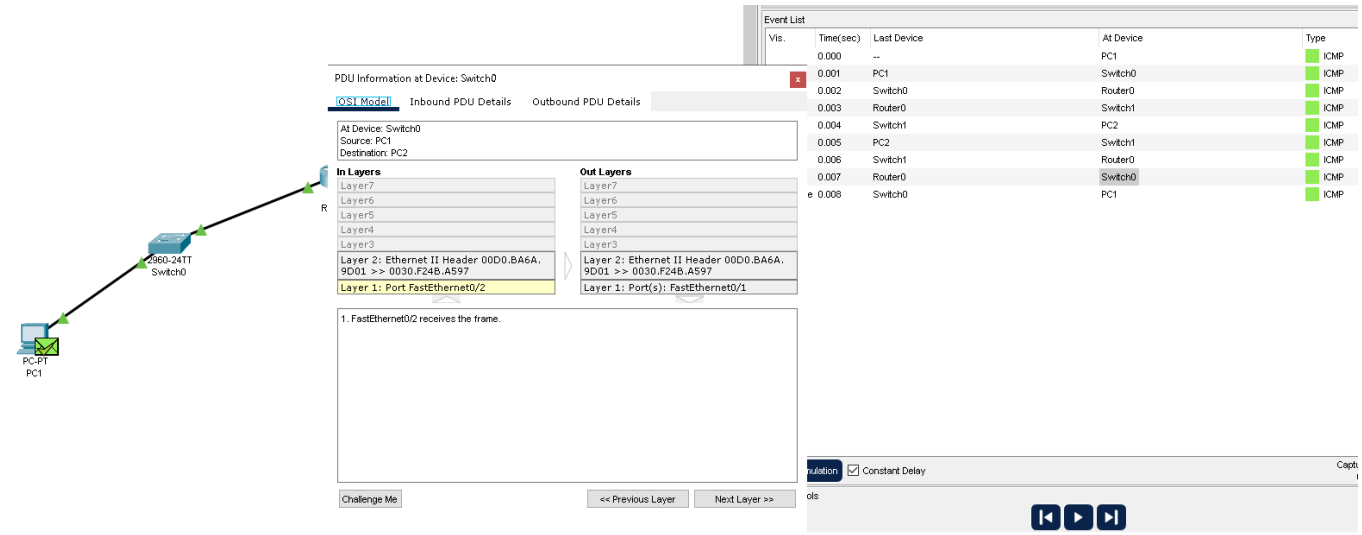
This step shows Switch1 receiving the return ICMP Echo Reply frame from PC2 through FastEthernet0/2. The switch, which operates at the Data Link Layer (Layer 2) of the OSI model, analyzes the Ethernet header to determine the source and destination MAC addresses. At this point, the process of decapsulation has started, as the network devices begin to remove the previously added headers to prepare the data for delivery back to the sender (PC1).

From Switch1 to Router:



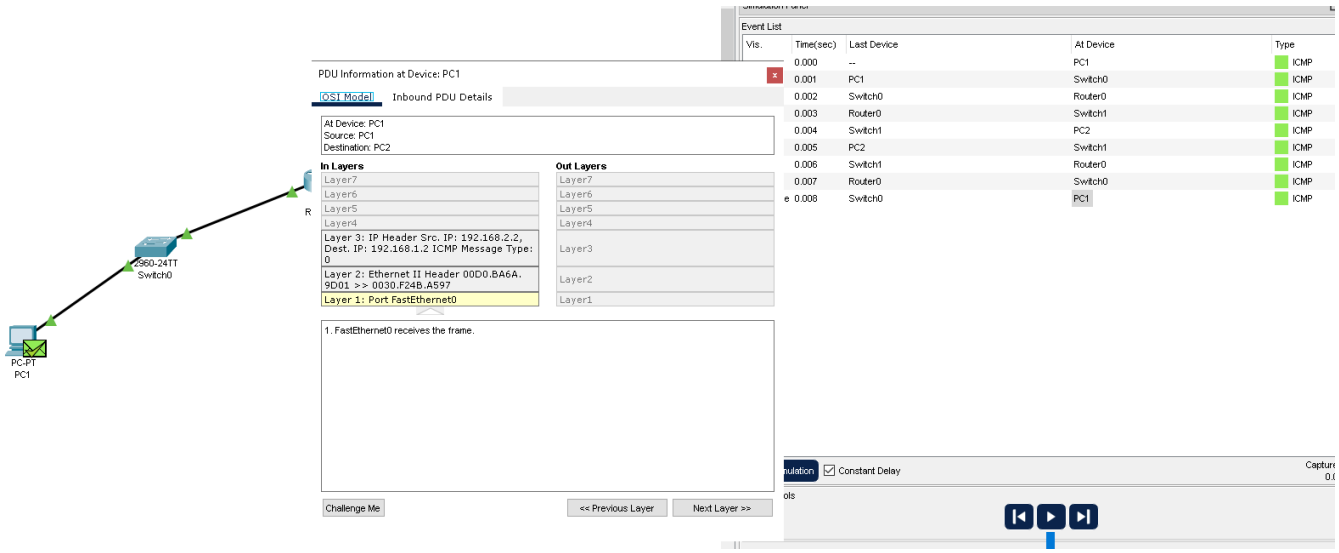
This step shows Router0 receiving the ICMP Echo Reply frame from Switch1 through GigabitEthernet0/1. The router, which operates at the Network Layer (Layer 3), inspects the IP header to determine the source IP (192.168.2.2) and the destination IP (192.168.1.2).

From Router to Switch0:



This step shows Switch0 receiving the ICMP Echo Request from PC1 through FastEthernet0/2. Operating at Layer 2 (Data Link Layer), the switch reads the Ethernet header, checks the MAC addresses, and forwards the frame out FastEthernet0/1 toward the destination.

From Switch0 to PC1:



This step shows PC1 receiving the ICMP Echo Reply from PC2 through FastEthernet0. At this point, the decapsulation process is completed as PC1 removes the Ethernet and IP headers to extract the original ICMP data. The message confirms that the communication between PC1 and PC2 was successful.